Information Technology Institute

ITI

Introduction to Computer Networks & Cyber Security

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Cyber Security Essentials

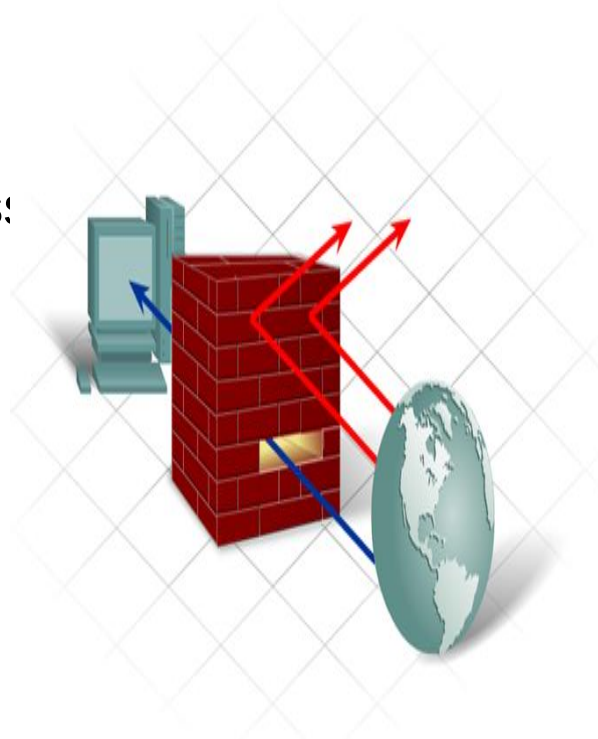
how to protect your Systems

Attack Mitigation

- **Hardware**
  - Firewalls
  - DMZ
  - IDS/IPS
  - NGFW
- **Software**
  - Anti-virus
  - Anti-spam
  - Anti-malware
  - Security Patches
  - User Access Control

Firewall

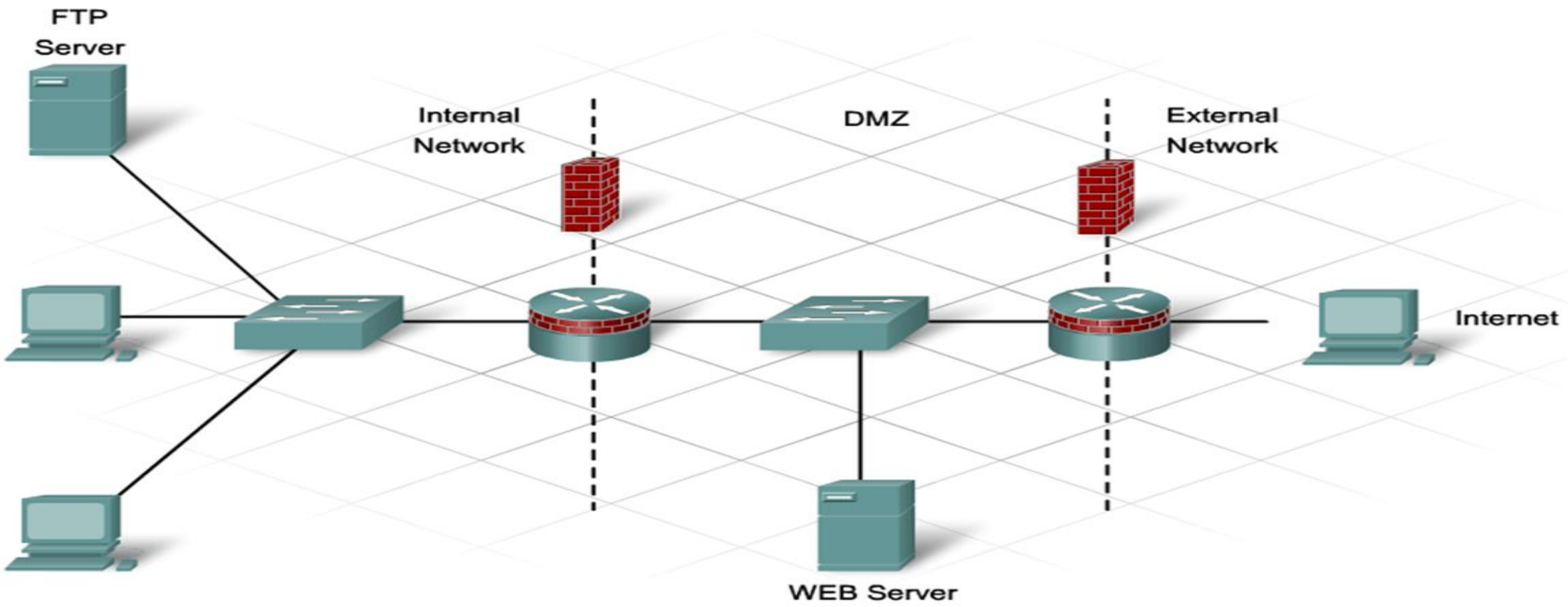
- A Firewall is one of the most effective security tools available for protecting internal network users from external threats As the first line of defense
- A firewall resides between two or more networks and controls the traffic between them as well as helps prevent unauthorized acces
- A firewall can be software-based or hardware-based



- **Static Packet Filtering (stateless firewall)**
  - - Prevents or allows access based on IP or MAC addresses.

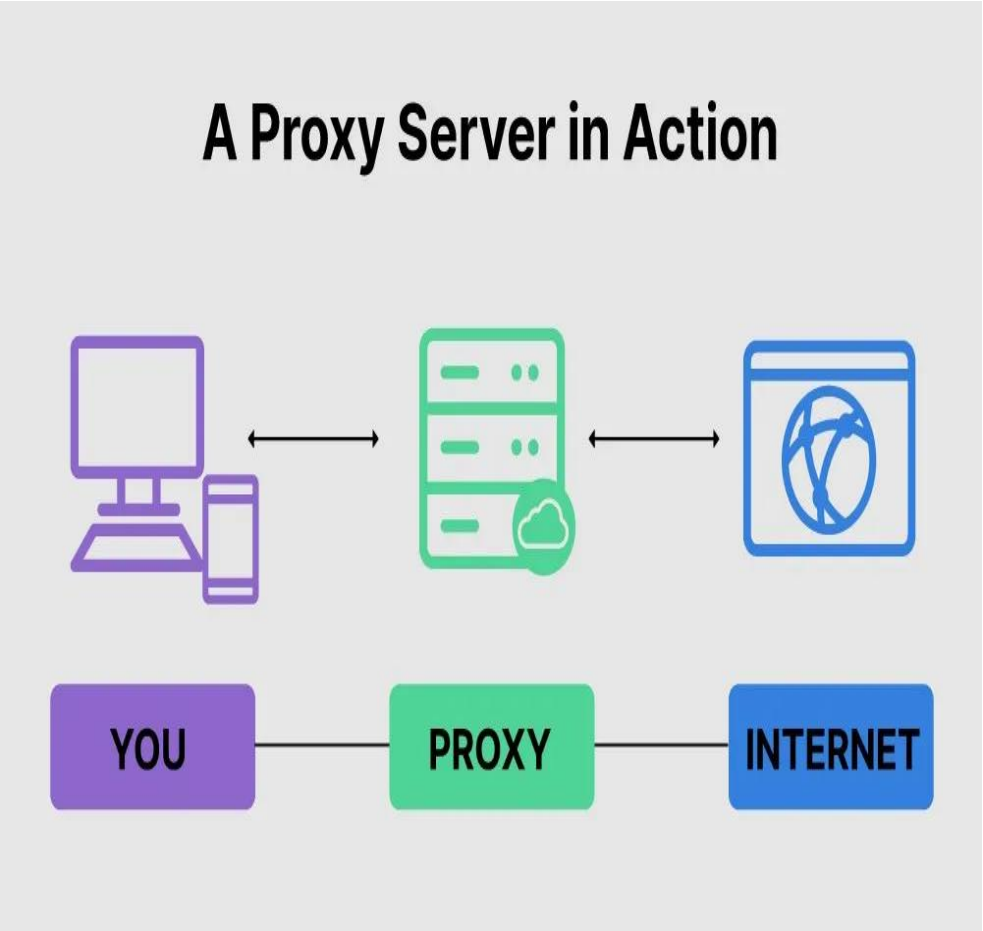
- **Dynamic Packet Filtering (state full firewall)**
  - Incoming packets must be legitimate responses to requests from internal hosts. filter out specific types of attacks such as DoS

DMZ


FTP
Server
Internal
Network
DMZ
External
Network
Internet
WEB Server

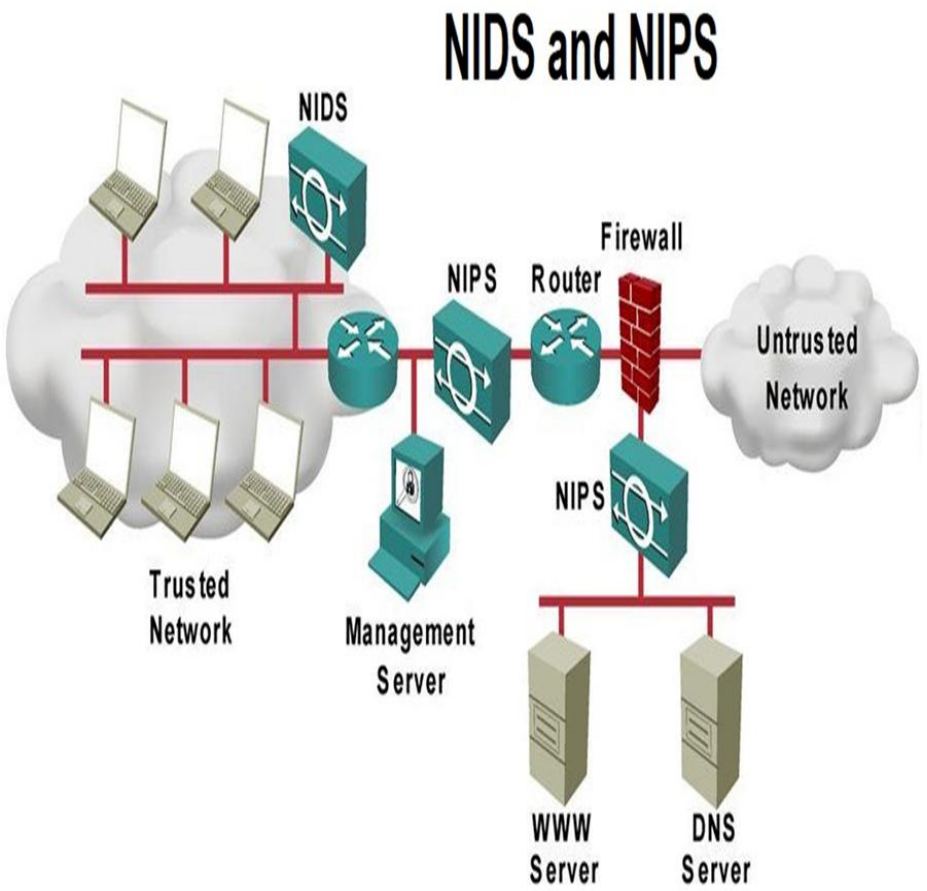
Proxy Server

- A **computer system** (or an application program) that intercepts internal user requests and then processes that request on behalf of the user
- Goal is to **hide the IP address** of client systems inside the secure network



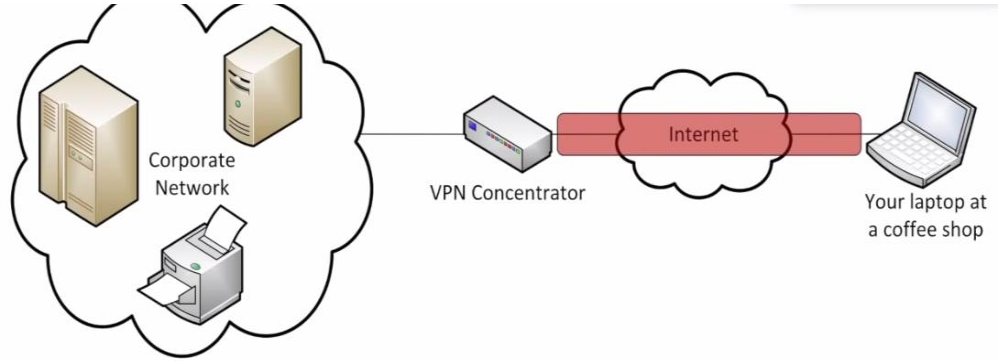
IDS/IPS

- Network Intrusion Detection System (NIDS):
  - **Watch** the Network Traffic and if there **is Intrusion it Detects** that there is Bad traffic Flow.
  - it **send alarms and logs**
- Network Intrusion prevention System (NIPS):
  - **Stops** the traffic if it **detects that there is intrusion**
- Types of IDS&IPS
  - **Signature-based:** look for the perfect match
  - **Anomaly-based:** Built a based line of what is normal
  - **Behavior-based:** **observe and report**



VPN

- It Tunnel the traffic between the Two Sides of Network
- Types:
  - Remote Access VPN
  - Site to Site VPN

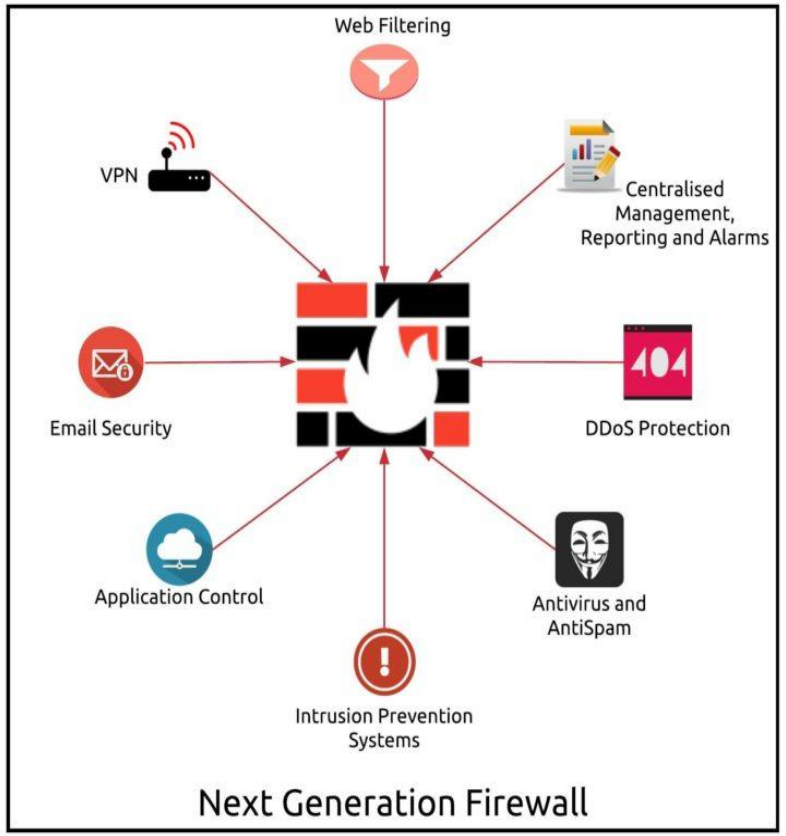




SIte-to-Site VPN

Next generation Firewall (NGFW)

- **Next generation Firewall (NGFW)**
  - a “deep-packet inspection firewall that moves beyond port/protocol inspection and blocking to add application-level inspection, intrusion prevention, and bringing intelligence from outside the firewall.”



Wireless Security

- Open Access
  - SSID
  - No encryption
  - Basic authentication
  - Not a security handle
- WPA2
  - AES Encryption
  - Authentication
- WPA3
  - Protected from Brute-force attack

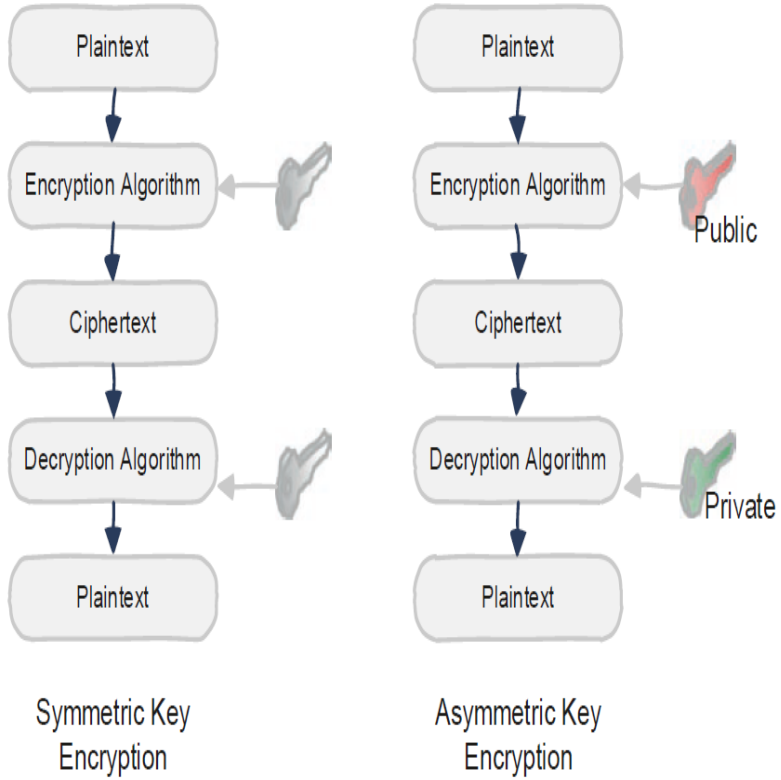


Controlling Wireless LAN Access

- SSID broadcasts from access points are off
- MAC Address filtering is enabled
- WPA2 / WPA3 Security implemented

Part 2_ Encryption

- **Encryption**
  - encryption is the process of **encoding information.** This process converts the original representation of the information, known as **plaintext**, into an alternative form known as **ciphertext.**
  - Unencrypted data, called plaintext, is sent through an encryption algorithm to generate a ciphertext. **A key** is used for encryption.
  - in **a symmetric encryption** algorithm, the **same key** is also **used for decryption**. (Not secure) needs to be a secure way for the two sides to have the same key

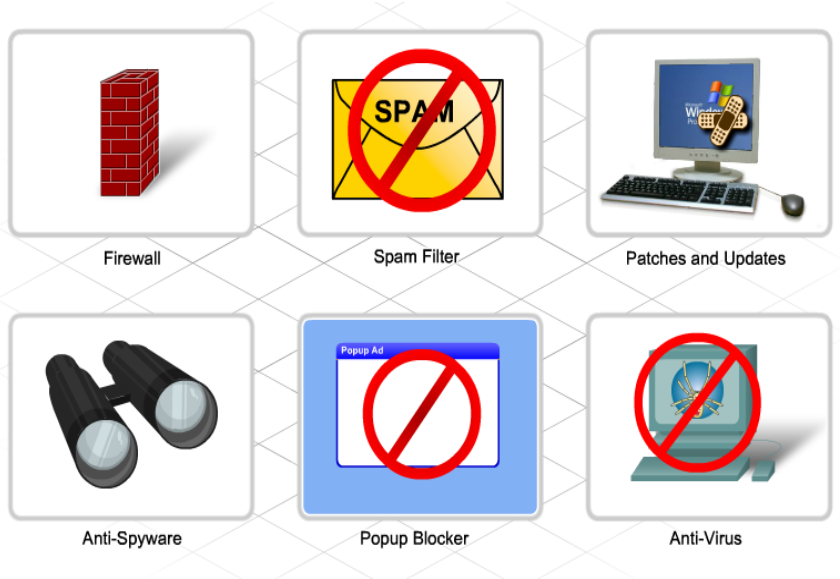


Firewall_ lab _Demo

- How to use your local firewall to block a port and stop DOS attack from a zombie device

Best Practices

- ✓ Define security policies
- ✓ Physically secure servers and network equipment
- ✓ Set login and file access permissions
- ✓ Update OS and applications
- ✓ Change permissive default settings
- ✓ Run anti-virus and anti-spyware
- ✓ Update antivirus software files
- ✓ Activate browser tools –
- ✓ Popup stoppers, anti-phishing, plug-in monitors
- ✓ Use a firewall



Thank You